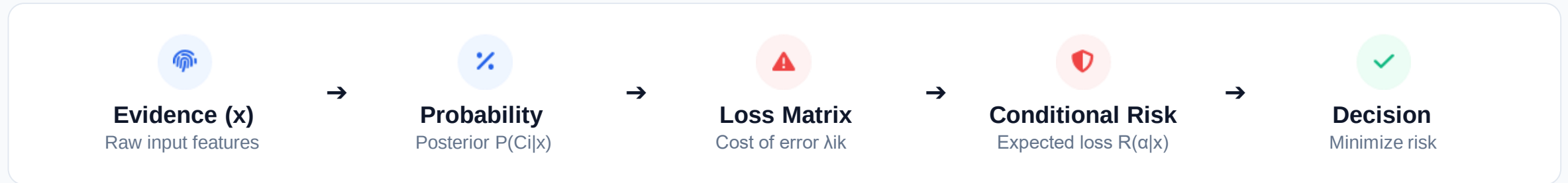


UNIT II — BAYESIAN DECISION THEORY • CLASS 2

# Discriminant Functions, Utility Theory & Value of Information

From scoring classifications and maximizing expected rewards to evaluating whether acquiring new features is worth the cost.

# Where We Stopped & Where We Go Today



## 1. Discriminant Functions

How computers score classes directly.

## 2. Utility Theory

Maximizing rewards instead of losses.

## 3. Value of Information

Deciding if extra data justifies cost.

**Teacher Note & Classroom Script:** Tell students: "Last time we proved that minimizing loss gives optimal decisions. But calculating loss every time in code is slow. Today we convert risks into scoring functions called discriminants, then explore utility and testing costs."

# What is a Discriminant Function?

## Core Textbook Definition

"A discriminant function is a mathematical function that assigns a numerical score to each class and chooses the class with the highest score."

**Input  $x$ :** Feature measurements from observation.

**Classes  $C_1, C_2, \dots, C_k$ :** The target categories.

**Scoring:** Every class receives its own score.

**Selection:** The highest score wins.

## Intuitive Working Diagram

Feature Input ( $x$ ): Image Pixels

▼ Passes through scoring functions

Score for Cat

**0.80**

Score for Dog

**0.20**

▼ Compare scores & select maximum

**Decision: Class Cat (Score 0.80 > 0.20)**

**Teacher Note & Classroom Script:** Explain as: "Think of an election where candidates get points based on voting ballots. The candidate with the maximum points is elected. A discriminant function simply tallies points for each category."

# Why Do We Need Discriminant Functions?



## Standardized Representation

Uniform framework for 2-class and K-class problems.



## Computational Efficiency

Avoids calculating full evidence denominators.



## Geometric Partitioning

Divides feature spaces into distinct decision regions.



## Key Computer Science Takeaway

A classifier is essentially a collection of K functions:  $g_1(x)$ ,  $g_2(x)$ , ...,  $g_K(x)$ .

`argmax_i [  $g_i(x)$  ]`

**Teacher Note & Classroom Script:** Tell students: "Why do programmers love this? In Python, instead of writing nested if-else statements with probability laws, we evaluate an array of formulas `[g1(x), g2(x)]` and return `np.argmax(g)`. It turns logic into arithmetic."

# The Discriminant Decision Rule

## FORMAL DECISION RULE

Choose  $C_i$  if  $g_i(x) = \max_k g_k(x)$

### Symbol Breakdown:

$x$  : The observed feature vector or sample.

$g_i(x)$  : Discriminant score computed for candidate class  $C_i$ .

$\max_k$  : Look across all classes  $k = 1, \dots, K$  and pick the largest.

**Plain English:** Calculate the score for each class; pick the highest.

## Quick Numerical Example

|                   |      |
|-------------------|------|
| Class A ( $g_A$ ) | 0.70 |
| Class B ( $g_B$ ) | 0.20 |
| Class C ( $g_C$ ) | 0.10 |

**Result:**  $\max(0.70, 0.20, 0.10) = 0.70 \rightarrow$  Class A is selected!

**Teacher Note & Exam Tip:** Reassure students: "Whenever you see  $\max_k$ , do not get intimidated by calculus notations. It literally just means 'find the maximum value in the list'."

# Relationship Between Risk and Discriminant Functions

## Previous Class Principle

### Minimum Risk Rule

Choose action  $\alpha_i$  with the lowest conditional risk:

$$\text{Select } \alpha_i = \operatorname{mink} R(\alpha_k | x)$$

## Today's Discriminant View

### Maximum Score Rule

Choose class  $C_i$  with the highest discriminant score:

$$\text{Select } C_i = \operatorname{maxk} g_k(x)$$

## FUNDAMENTAL EQUIVALENCE FORMULA

$$g_i(x) = -R(\alpha_i | x)$$

**Why the negative sign?** Minimizing a positive loss is identical to maximizing its negative value:  $\min(0.20) \rightarrow \max(-0.20)$ .

**Exam Core Rule:** Maximum discriminant score corresponds directly to minimum conditional risk!

**Teacher Note & Classroom Script:** Ask students: "Which temperature is better in winter:  $-2^\circ\text{C}$  or  $-10^\circ\text{C}$ ?  $-2^\circ\text{C}$  is higher! Similarly, a loss of 0.2 gives score  $-0.2$ , while loss of 0.7 gives  $-0.7$ .  $-0.2 > -0.7$ , so maximum score correctly minimizes risk."

# Discriminant Functions Under 0–1 Loss

## What is 0–1 Loss?

- **Correct Decision: Loss = 0 (No penalty)**
- **Wrong Decision: Loss = 1 (Equal penalty for any mistake)**

Under this standard assumption, risk simplifies directly to the probability of making an error:  $R(\alpha_i | x) = 1 - P(C_i | x)$ .

### SIMPLEST DISCRIMINANT FUNCTION

$$g_i(x) = P(C_i | x)$$

## Posterior as the Score

When mistakes cost equally, the posterior probability itself acts directly as the discriminant score.

**P(Cat | x)** **0.80**

**P(Dog | x)** **0.20**

**Choose Cat because  $0.80 > 0.20$**

**Teacher Note & Exam Tip:** Students frequently get asked: "What is the discriminant function under 0-1 loss?" Tell them the answer is simply: "The posterior probability  $P(C_i|x)$ ". Ethem Alpaydin highlights this in Chapter 3.

# Decision Regions & Decision Boundary

## Two Key Definitions

### Decision Region ( $R_i$ ):

The entire area in feature space where class  $C_i$  has the highest score.

### Decision Boundary:

The line or surface where two classes have equal scores:  
 $g_1(x) = g_2(x)$ .

On the boundary, the classifier is equally undecided between classes.



Region R1 (Class A)

Boundary:  $g_1(x) = g_2(x)$

Region R2 (Class B)

**Teacher Note & Classroom Script:** Draw a line on the board dividing the classroom: "Everyone on the left side is Class A; everyone on the right is Class B. The line in the middle where both scores tie is the Decision Boundary."

# Discriminant Functions: Quick Recap

| Concept               | Meaning & Plain-English Interpretation                   | Key Formula               |
|-----------------------|----------------------------------------------------------|---------------------------|
| Discriminant Function | Assigns a numerical score to a given class for input $x$ | $g_i(x)$                  |
| Decision Rule         | Always pick the class that scored the absolute highest   | Choose $\max_k g_k(x)$    |
| Connection to Risk    | Maximizing score equals minimizing conditional risk      | $g_i(x) = -R(\alpha_i x)$ |
| Under 0–1 Loss        | Posterior class probability is itself the discriminant   | $g_i(x) = P(C_i x)$       |

Memory Mantra: "Score Each Class → Compare Scores → Choose the Highest"

**Teacher Note & Exam Tip:** This table is a guaranteed 5-mark answer if students get asked: "Write a short note on Discriminant Functions."

# Introduction to Utility Theory

Loss = Penalty

Utility = Reward

**"Utility is a numerical measure of the benefit, value, satisfaction, or desirability of an outcome."**

In many real-world machine learning systems (finance, games, recommendation engines), people don't think in terms of penalties—they think in terms of **gains, profit, and health improvements**.

## Real-World Utility Scale

**High Profit / Patient Cured** High Positive Utility (+100)

**Break-Even / Minor Symptom** Moderate Utility (+10)

**Severe Financial Loss / Harm** Zero or Negative Utility (-50)

**Teacher Note & Classroom Script:** Ask students: "When you play a mobile game, do you try to minimize your damage, or maximize your gold coins? They are two sides of the same coin! Utility theory is about maximizing coins."

# Utility vs. Loss: The Fundamental Duality

## LOSS ( $\lambda$ )

Cost / Penalty

- Measures cost, damage, penalty.
- Higher values are worse.
- Used in classical statistical risk.

**Goal: MINIMIZE Loss**

## UTILITY (U)

Gain / Value

- + Measures benefit, profit, desirability.
- + Higher values are better.
- + Used in AI decision systems & economics.

**Goal: MAXIMIZE Utility**

**Teacher Note & Classroom Script:** "Stress this table heavily! In university question papers, 'Differentiate between Loss and Utility' is a standard 5-mark question. Just remember: Loss = Minimize, Utility = Maximize."

# How Utility Drives Machine Decisions

## 1. Actions ( $\alpha_i$ )

Decisions we can take (e.g.,  
Treat patient or Discharge).

## 2. States ( $S_k$ )

True condition of nature (e.g.,  
Patient is Sick or Healthy).

## 3. Probabilities

Belief given evidence:  $P(S_k | x)$ .

## 4. Utility ( $U_{ik}$ )

Payoff of choosing action  $\alpha_i$   
when state  $S_k$  happens.

### THE RATIONAL DECISION PIPELINE

Observe Evidence ( $x$ ) → Compute  $P(S_k|x)$  → Weigh by Utilities  $U_{ik}$  → **Max Expected Utility**

**Teacher Note & Classroom Script:** "Explain: We don't know the exact future state with 100% certainty. But we know the probabilities and the payoffs. Expected Utility combines both into a single scorecard."

# Expected Utility: The Master Formula

## EXPECTED UTILITY OF ACTION $A_i$

$$EU(\alpha_i | x) = \sum_k U_{ik} \cdot P(S_k | x)$$

### Variable-by-Variable Explanation:

$EU(\alpha_i | x)$  : The average payoff if we execute action  $\alpha_i$ .

$U_{ik}$  : Utility value obtained if action  $\alpha_i$  is taken and state  $S_k$  is true.

$P(S_k | x)$  : Probability that the true state is indeed  $S_k$ .

$\sum_k$  : Sum the products over all possible states  $k = 1, \dots, K$ .

## The Translation Rule

"Expected Utility = Sum of (Utility  $\times$  Probability) for every possible state."

**Decision Rule:** Always pick the action with maximum expected reward.  
Select  $\alpha^* = \operatorname{argmax}_i EU(\alpha_i | x)$

**Teacher Note & Classroom Script:** "Reiterate to students:  $\Sigma$  is just a loop! In programming: ``total = sum([U[i][k] * P[k] for k in states])``. That is all this mathematical formula represents."

# Expected Utility in Action: Medical Diagnosis

## Given Information

$P(\text{Sick}) = 0.60$ ,  $P(\text{Healthy}) = 0.40$

Utilities for Action 1 (Treat):  
 $U(\text{Treat, Sick}) = 10$  units

$U(\text{Treat, Healthy}) = 2$  units

## Step-by-Step Evaluation

$$\begin{aligned} EU(\text{Treat}) &= (10 \times 0.60) + (2 \times 0.40) \\ &= 6.0 + 0.8 \\ &= \mathbf{6.8 \text{ units}} \end{aligned}$$

$$EU(\text{Do Not Treat}) = (0 \times 0.60) + (6 \times 0.40) = \mathbf{2.4 \text{ units}}$$

**Decision: Since  $6.8 > 2.4$ , choose action "Treat"!**

**Teacher Note & Classroom Script:** "Point out: Even though giving medicine to a healthy person has low utility (2), the high utility of curing the sick person (10) combined with 60% probability makes 'Treat' the overwhelmingly rational choice."

# Risk vs. Expected Utility: Two Paths to Rationality

| Dimension           | Conditional Risk $R(\alpha x)$                                   | Expected Utility $EU(\alpha x)$          |
|---------------------|------------------------------------------------------------------|------------------------------------------|
| Underlying Focus    | Expected Loss (Penalties / Costs)                                | Expected Benefit (Gains / Payoffs)       |
| Optimization Target | MINIMIZE Risk                                                    | MAXIMIZE Utility                         |
| Mathematical Sign   | Positive costs to be avoided                                     | $U = -\text{Loss}$ (Inverting the scale) |
| Core Equivalence    | Minimum Expected Risk $\Leftrightarrow$ Maximum Expected Utility |                                          |

Takeaway: Both frameworks produce the exact same optimal decision in every machine learning problem.

Teacher Note & Exam Tip: "Tell students: If an exam question asks 'Is Expected Utility incompatible with Bayesian Risk?', the answer is NO! They are mathematical mirrors of each other."

# Should We Collect More Information?

## The Core Dilemma

**"Value of Information (VOI) measures how much the quality of a decision improves if we gather additional features before acting."**

Features are not free. In medicine, tests cost money and time. In machine learning, computing features requires sensor hardware and processing power.

## Real-World Medical Example

### Scenario A: Act Immediately

Doctor observes cough & fever. Decides based on limited evidence.

### Scenario B: Order Blood Test

Doctor delays decision, runs a test. Gains clearer diagnosis.

**Key Question: Does the test justify its cost?**

**Teacher Note & Classroom Script:** "Ask students: 'If an MRI test costs ₹10,000 to detect a minor headache, would you order it?' No! But for suspected trauma, yes. Value of Information is the math behind that clinical judgment."

# Existing Information (x) vs. New Information (z)

## **x** Existing Evidence

Information already known at zero extra cost.

- Patient age, gender, body temperature
- Email subject line, sender domain
- Current stock market opening index

## **z** Additional Feature

New feature that can be acquired for a price.

- Specialized blood test or biopsy
- Deep virus scan / sandbox execution
- Expensive paid market research survey

## Updated Knowledge State: (x, z)

Combines prior symptoms with test results to refine posterior probability.

$$P(S_k | x) \rightarrow P(S_k | x, z)$$

**Teacher Note & Classroom Script:** "Make sure students note down `x` and `z`. In exams, students often forget which letter stands for existing features and which stands for the new test."

# Quantifying Value of Information (VOI)

## VALUE OF INFORMATION EQUATION

$$\text{VOI} = \text{EU}(x, z) - \text{EU}(x)$$

### Component Meanings:

**EU(x)** : Best expected utility achievable using *only current information x*.

**EU(x, z)** : Improved expected utility achieved *after learning extra feature z*.

**VOI** : The net gain in expected decision quality from the new data.

### Plain-English Translation

"Value of Information is the improvement in expected reward when we make a decision with more facts."

### Theoretical Property:

VOI is always  $\geq 0$ . Additional data cannot harm decision quality; it can only maintain or improve it.

**Teacher Note & Exam Tip:** "Ask students: 'Can VOI be negative?' Explain why it cannot: If the new test tells you nothing useful, you simply stick with your original decision, so  $\text{EU}(x, z) = \text{EU}(x)$ , making  $\text{VOI} = 0$ . It is never negative."

# VOI Numerical Example: Test vs. Cost

## Decision States

Current Expected Utility

$$EU(x) = 60 \text{ units}$$

Expected Utility with Blood Test

$$EU(x, z) = 85 \text{ units}$$

## 1. Gross Value of Information

$$VOI = EU(x, z) - EU(x) = 85 - 60 = 25 \text{ units}$$

## 2. Deduct Cost of Test

Assume the laboratory charges: **Cost(z) = 10 units.**

$$\text{Net Benefit} = VOI - \text{Cost} = 25 - 10 = +15 \text{ units}$$

**Conclusion: Net benefit is positive (+15). YES, the test is worth taking!**

**Teacher Note & Classroom Script:** "Now ask: What if the test cost was 30 units? Net benefit would be  $25 - 30 = -5$ . In that case, we should decline the test and decide immediately! Always check net benefit."

# When is Information Worth Getting?

✓ **VOI > Cost of Information**

**Action: GATHER FEATURE  $z$**

The expected improvement in outcome outweighs the financial, computational, or time expense of acquiring the data.

✗ **VOI  $\leq$  Cost of Information**

**Action: DO NOT GATHER (ACT NOW)**

The test is overpriced relative to the clarity it offers. The algorithm should make the best possible call with existing data  $x$ .

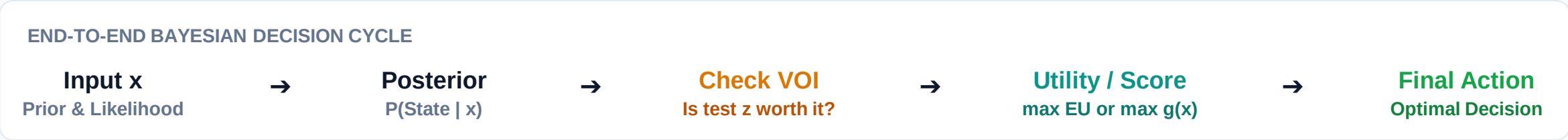
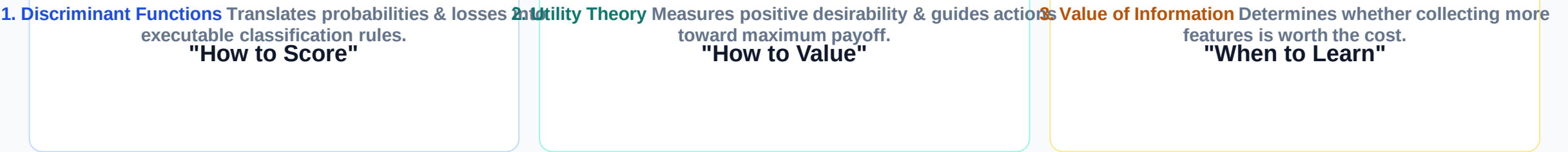
## Essential Statistical Clarification:



VOI evaluates *expected* benefit across all possible test outcomes; it does not guarantee that every individual test will change the final decision.

**Teacher Note & Classroom Script:** "Give an everyday example: If an online course charges ₹50,000 but only increases your expected starting salary by ₹5,000,  $VOI < \text{Cost}$ ! Rational decision theory applies to life decisions."

# How Today's Three Topics Fit Together



**Teacher Note & Classroom Script:** "Summarize: Machine learning isn't just fitting a line. It is a complete pipeline of observing, testing if more data is needed, evaluating payoffs, and picking the winning action."

# Key Formulas & Rules for BCA Exams

## Formulas You MUST Memorize

Risk Equivalence

$$g_i(x) = -R(\alpha_i | x)$$

0-1 Loss Discriminant

$$g_i(x) = P(C_i | x)$$

Expected Utility

$$EU(\alpha_i | x) = \sum_k U_{ik} \cdot P(S_k | x)$$

Value of Information

$$VOI = EU(x, z) - EU(x)$$

## Core Decision Criteria

→ **Discriminant:** Pick max  $g_i(x)$

→ **Bayesian Risk:** Pick min  $R(\alpha_i | x)$

→ **Expected Utility:** Pick max  $EU(\alpha_i | x)$

→ **Acquire Data:** If  $VOI > \text{Cost}$

**Direct Match for Alpaydin Ch 3 Questions!**

**Teacher Note & Exam Tip:** "Tell the class to take a screenshot or write this slide down in their revision notebook. 80% of Unit II numerical and short-answer questions are solved with these 4 lines."

# Model University Exam Questions & Hints

## 2 MARKS (Definitions)

**Q: Define discriminant function.**

Hint: Score-assigning function; choose max.

**Q: Define VOI.**

Hint: Difference between  $EU(x,z)$  and  $EU(x)$ .

**Q: What is a decision boundary?**

Hint: Surface where  $g_1(x) = g_2(x)$ .

## 5 MARKS (Analytical)

**Q: Differentiate Loss and Utility.**

Hint: Loss = penalty/min; Utility = reward/max.

**Q: Explain Expected Utility with an example.**

Hint: Formula + medical Treat/No-treat example.

**Q: When is additional information worth getting?**

Hint: Explain  $VOI > \text{Cost trade-off}$ .

## 10 MARKS (Essay)

**Q: Explain Discriminant Functions & their relationship with Bayesian Risk.**

Hint: Define  $g_i(x)$ , derivation under 0-1 loss, decision boundary diagram.

**Q: Discuss Utility Theory and Value of Information in ML decision making.**

Hint: EU formula,  $x$  vs  $z$ , VOI formula & cost analysis.

**Teacher Note & Classroom Script:** "Advise students: In a 10-mark question, always draw the decision boundary diagram and write out the numerical example we covered. That ensures full marks from external examiners."

# The Three Pillars of Rational Decisions

## 1. Discriminants

Choose the highest score.  
Give classes a score.

## 2. Utility Theory

Maximize expected utility.  
Measures benefits.

## 3. Value of Info

Collect if  $VOI > \text{Cost}$ .  
Evaluates new tests.

"Bayesian Decision Theory helps intelligent systems make optimal choices in an uncertain world."



Syllabus Topics 100% Covered



Open for Student Questions & Discussion

**Teacher Note:** Congratulate the students on completing the mathematical core of Unit II. Preview next lecture on Bayesian Networks (directed acyclic graphs) and influence diagrams.